

## 1. Practical 7 — Memory Layout

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

int main()
{
    int local_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("Process ID: %d\n", getpid());

    printf("Code address : %p\n", (void *)main);
    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address : %p\n", (void *)heap_var);
    printf("Stack address : %p\n", (void *)&local_var);

    printf("\nProcess will remain active for 30 seconds...\n");

    sleep(30);

    free(heap_var);

    return 0;
}

hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ cd ~/OSSP_Assignment
hemasai@Hemasai:~/OSSP_Assignment$ mkdir -p Practical_7_8
hemasai@Hemasai:~/OSSP_Assignment$ cd Practical_7_8
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ nano memory_layout.c
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ gcc memory_layout.c -o memory_layout
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ ./memory_layout
Process ID: 1982
Code address : 0x5705d96cc209
Global address : 0x5705d96cf010
Static address : 0x5705d96cf014
Heap address : 0x5705e85c2010
Stack address : 0x7fffe0a4751c

Process will remain active for 30 seconds...
```

## 2. Practical 8(a) — Dynamic Memory Allocation

```

GNU nano 2.7.1
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    a = (int *)malloc(5 * sizeof(int));

    b = (int *)calloc(5, sizeof(int));

    a = (int *)realloc(a, 10 * sizeof(int));

    free(a);
    free(b);

    return 0;
}

```

```

Process will remain active for 30 seconds...
^C
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ nano memory_allocation.c
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ gcc memory_allocation.c -o memory_allocation
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ ./memory_allocation
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ valgrind --leak-check=full ./memory_allocation
==2032== Memcheck, a memory error detector
==2032== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2032== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2032== Command: ./memory_allocation
==2032==
==2032== HEAP SUMMARY:
==2032==    in use at exit: 0 bytes in 0 blocks
==2032==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==2032==
==2032== All heap blocks were freed -- no leaks are possible
==2032==
==2032== For lists of detected and suppressed errors, rerun with: -s
==2032== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$

```

### 3. Practical 8(b) — Copy-on-Write

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before = %d\n", x);

        x = 500;

        printf("Child After = %d\n", x);
    }
    else
    {
        sleep(2);

        printf("Parent = %d\n", x);
    }

    return 0;
}

```

```

==2032==
==2032== For lists of detected and suppressed errors, rerun with: -s
==2032== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ nano cow.c
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ gcc cow.c -o cow
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ ./cow
Child Before = 100
Child After = 500
Parent = 100
hemasai@Hemasai:~/OSSP_Assignment/Practical_7_8$ |

```